

Revised Generation Interconnection System Impact Study Report

For Queue Project Q19-1159

Olivehurst 100 kV
May 26, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1159. The Interconnection Customer proposes a 64.9 MW wind facility interconnecting at the Olivehurst 100 kV substation in the Ohio area. The request was received on March 04, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 418 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$3,796,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Sierra Linda 174-Ephrata 100 kV	Eighteenth Street-Lagoon Creek 100 kV Line	113.9	126	45.9
Eighteenth St-Lagoon Creek 100 kV Line	Ephrata-Olivehurst 100 kV L	103.8	192	61.4
Ephrata-Olivehurst 100 kV	Eighteenth St-Lagoon Crk 100 kV	101.4	158	45.9

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Sierra Linda 1704-Ephrata 100 kV Line	Rebuild line with higher-capacity conductor	\$10,730,000	23% (\$2,500,000)
Eighteenth St-Lagoon Creek 100 kV Line	Reconductor line	\$2,140,000	22% (\$459,000)
Ephrata-Olivehurst 100 kV	Reconductor line	\$3,591,000	23% (\$837,000)
	Total allocated to project		\$3,796,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.